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Programming for Artificial Intelligence and Data Science (CO3135)

Assignment Submission

PREDICTING REAL ESTATE ASKING PRICES IN VIETNAM

A Study of Machine Learning and Deep Learning Approaches

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1 Group Information & Member Roles

The group consists of five members working on real estate asking-price prediction in Vietnam. Responsibilities are organised into dataset preparation and exploratory data analysis, machine learning, and deep learning. Table 1 lists the individual contributions.

Table 1: Group members and assigned responsibilities.

Member / Student ID	Area	Responsibilities
Nguyễn Mạnh Quốc Khánh 2352525	Dataset and EDA	Dataset selection and preparation; schema and missing-value analysis; duplicate and outlier handling; exploratory analysis of price, area, location, and property type; preparation of the dataset and EDA report.
Võ Nhật Tân 2353081	Machine learning	Structured Ridge regression, Random Forest, LightGBM, and CatBoost; model-specific preprocessing, training, validation, and interpretation of the assigned models.
Phan Ngọc Lan Chi 2352137	Machine learning	XGBoost, text-only TF-IDF Ridge, joint text-and-structured Ridge, and LightGBM with a text-derived prediction feature; text representation, model training, validation, and interpretation of the assigned models.
Phạm Thanh Bảo Ngân 2352797	Deep learning	Tabular multilayer perceptron with categorical embeddings; numerical preprocessing, embedding construction, network implementation, optimisation, and early stopping.
Lê Phước Minh Thiện 2353132	Deep learning	BamiBERT text regression; text preparation, tokenisation, target standardisation, pretrained-model fine-tuning, and checkpoint evaluation.

Shared machine learning responsibilities. Võ Nhật Tân and Phan Ngọc Lan Chi jointly establish the constant-median and local median-price-per-square-metre baselines, prepare the shared feature representation and chronological evaluation protocol, compare the machine learning models, and write the machine learning report. The text-only TF-IDF Ridge model also supplies the prediction feature used by the text-enhanced LightGBM.

Shared deep learning responsibilities. Phạm Thanh Bảo Ngân and Lê Phước Minh Thiện are each responsible for one principal neural model. They jointly compare the deep learning results, implement and evaluate the prediction blends combining structured and text-based models, and write the deep learning report.

2 Chosen Data & Justification

The chosen dataset provides a large number of real-estate listings with a clear numeric target and a wide range of predictors, which suits the prediction of real estate asking prices in Vietnam.

Dataset *Tinix Vietnam Real Estate Listings 2025–2026* [Tin26]

Overview The collection contains 3,500,744 sale and rental listings published between June 2025 and March 2026, after a strict property-type filter, and covers all 63 provinces and cities of Vietnam. The listings are distributed as ten Parquet shards, are written primarily in Vietnamese, and are released under the CC BY-NC 4.0 licence for non-commercial academic use.

Justification The dataset contains over 3.5 million records, which supports stable exploratory analysis and supervised learning across regions and property types. Its schema separates a numeric target (`price`) from predictors describing location, property category, size and layout, orientation, textual content, and listing time. Coverage across provinces supports the analysis of regional price differences. Publication timestamps allow training on older listings and evaluating predictions on newer listings, matching the temporal direction of the intended prediction task.

Limitations Some asking prices are undisclosed, structural attributes are frequently incomplete, and the collection includes both sale and rental advertisements without a variable that distinguishes them. These issues are examined in Section 4, where an analytical subset of priced listings is constructed for subsequent analysis and model training.

Table 2 defines the 19 variables used in the study.

Table 2: Variables in the real estate listing dataset [Tin26].

Column	Type	Description
`name`	string	Listing title (HTML stripped, phone numbers redacted)
`description`	string	Full listing description (HTML stripped, phone numbers redacted)
`property_type_name`	string	Property category (e.g., Căn hộ chung cư, Nhà, Đất, ...)
`province_name`	string	Province / City name (63 provinces)
`district_name`	string	District (Quận / Huyện)
`ward_name`	string	Ward (Phường / Xã)
`street_name`	string	Street name
`project_name`	string	Real estate development project name (if applicable)
`price`	float64	Listed price in VND (Vietnamese Dong). `null` if not disclosed.
`area`	float64	Total floor/land area in m^2
`floor_count`	float64	Number of floors in the building
`frontage_width`	float64	Frontage width in meters
`house_depth`	float64	Depth of the property in meters
`road_width`	float64	Width of the road in front of the property (meters)
`bedroom_count`	float64	Number of bedrooms
`bathroom_count`	float64	Number of bathrooms
`house_direction`	string	Cardinal facing direction (Đông, Tây, Nam, Bắc, Đông-Bắc, ...)
`balcony_direction`	string	Cardinal facing direction of the balcony
`published_at`	string	ISO 8601 datetime the listing was published

3 Work Progress Overview

The project developed through weekly milestones between 31 August and 11 November 2026. Table 3 summarises the deliverables reached at each stage, the main difficulties, and the corresponding adjustments. Khánh led dataset preparation and EDA; Tân and Chi developed the machine learning models; Phạm Thanh Bảo Ngân and Lê Phước Minh Thiện developed the neural models and prediction blends.

Table 3: Weekly progress, difficulties, and responses.

Week	Progress reached	Difficulty and response
1 31 Aug.–6 Sep.	Agreed on asking-price prediction, selected the Vietnamese real estate dataset, reviewed the available attributes, and divided the work among five members.	Listings include both sale and rental advertisements without a reliable transaction-type label. The group defined the target as advertised asking price and retained this ambiguity as a limitation.
2 7–13 Sep.	Completed initial loading, schema checks, data-type inspection, and missing-value summaries. Established a reusable local data copy for subsequent analysis.	The dataset contains approximately 3.5 million listings, making repeated loading costly. Local Parquet storage and categorical conversion reduced repeated data-access and memory overhead.
3 14–20 Sep.	Completed duplicate detection and checks for invalid price and area values. Produced missing-value summaries by property type.	Many structural attributes are absent, and missingness differs by property type. The group avoided dropping every incomplete listing and retained missingness for model-specific treatment.
4 21–27 Sep.	Completed the main price and area analyses, reviewed extreme observations, and established the cleaned modelling population of 2,397,001 listings.	Extreme prices, implausible dimensions, and invalid timestamps affected modelling suitability. Price, area, and price-per-area restrictions were applied; invalid optional dimensions were set to missing.
5 28 Sep.–4 Oct.	Completed hierarchical location keys, categorical missing-value handling, text preparation, and the shared chronological 70/15/15 split.	Repeated district names could conflate locations, while advertisements could disclose the target price. Parent-location prefixes and monetary-clause/digit removal addressed these issues, with residual text leakage retained as a limitation.
6 5–11 Oct.	Established constant-median and local price-per-square-metre baselines. Completed structured Ridge training and the three-value regularisation search.	Linear fitting required consistent treatment of skewed numeric inputs, missing values, and numerous categorical levels. Training-fitted log transforms, imputation, scaling, and rare-category pooling formed a common pipeline.
7 12–18 Oct.	Completed the Random Forest depth/feature-fraction comparison and the LightGBM, CatBoost, and XGBoost training configurations. Prepared validation metrics and feature-importance diagnostics.	Tree experiments on the full training population were computationally demanding. A 10% bootstrap fraction per forest tree, parallel CPU execution, and validation-based stopping controls limited the training workload.
8 19–25 Oct.	Completed TF-IDF features, joint text-and-structured Ridge, and the text-prediction input for LightGBM. Completed the tabular MLP architecture and its preprocessing pipeline.	Dense text matrices would be costly, and in-sample text predictions could mislead the downstream model. Sparse matrices and five-fold out-of-fold Ridge predictions were used; categorical embeddings reduced the neural input expansion.

Continued on the next page.

Table 3 continued.

Week	Progress reached	Difficulty and response
9 26 Oct.–1 Nov.	Developed the MLP optimisation and early-stopping procedure. Prepared BamiBERT tokenisation, standardised log-price targets, and GPU fine-tuning.	Neural training required control of sequence length, numerical scale, and GPU workload. Text was limited to 512 tokens, targets were standardised for BamiBERT, and mixed precision was configured.
10 2–8 Nov.	Consolidated neural-model predictions, compared structured and text-based approaches, and developed nonnegative prediction blends with validation-fitted coefficients.	Combining separately trained models required aligned observations and consistent prediction units. Validation/test predictions retained common row identities and log-price units. Separate reconstruction of text-model ordering remained a consistency check.
11 9–11 Nov.	Brought together experiment summaries, diagnostic figures, and report contributions. Reviewed model settings, interpretation, and consistency between individual and combined predictions.	Results from different experimental populations could not be pooled directly. Comparisons were restricted to compatible runs; remaining text-filtering and checkpoint-compatibility limitations were retained for discussion.

Fortnightly tracking and discussion. Following the kickoff on 31 August, the tracking milestones were set two weeks apart: 14 September (data loading and quality issues), 28 September (cleaning decisions and feature preparation), 12 October (baselines and evaluation protocol), 26 October (tree/text results and neural-training requirements), and 9 November (prediction alignment, model combinations, and report integration). At each checkpoint, members reviewed deliverables, discussed unresolved issues, and agreed on the next tasks. A final review on 11 November covered the consolidated findings and outstanding limitations. Intermediate outputs were also exchanged between meetings so that modelling work could proceed in parallel.

4 Exploratory Data Analysis (EDA)

Data-quality checks establish which listings can support price prediction and which attributes require special treatment. The analysis covers listings with a disclosed asking price in VND. Sale and rental advertisements cannot be separated reliably because the data do not contain a transaction-type label (Section 2).

4.1 Data Quality Analysis

4.1.1 Schema and Data Types

The raw data contain 3,500,744 listings and the 19 variables described in Table 2. The schema check confirmed that all expected columns are present and that there are no additional columns. The data types are as follows:

- Numeric (8 variables): `price`, `area`, `floor_count`, `frontage_width`, `house_depth`, `road_width`, `bedroom_count`, `bathroom_count`.
- Text (11 variables): all remaining variables.

To reduce memory usage, six categorical columns were converted to the categorical type, which reduced the memory footprint from 6.533 GB to 5.334 GB. A conservative cleaning strategy was adopted: quality issues were flagged and all rows were kept in the full dataset, while a separate subset of priced listings was built afterwards (Section 4.2).

4.1.2 Missing-Value Analysis

The null-value scan showed missing entries in 14 of the 19 variables (Table 4). The main findings are:

- `price`: 218,494 missing values (6.24%), and `area`: 4 missing values.
- Missingness is particularly high for direction and dimension attributes: 68.96% for `house_direction` and 98.04% for `house_depth`.
- `name`, `description`, `property_type_name`, `province_name`, and `published_at` have no missing values.

Table 4: Missing values by variable in the raw data ($N = 3{,}500{,}744$).

Variable	Missing (n)	Missing (%)
`house_depth`	3,432,288	98.04
`balcony_direction`	2,933,342	83.79
`road_width`	2,830,873	80.86
`floor_count`	2,691,982	76.90
`project_name`	2,499,891	71.41
`house_direction`	2,414,214	68.96
`frontage_width`	1,726,424	49.32
`bathroom_count`	1,720,974	49.16
`bedroom_count`	1,600,361	45.71
`street_name`	1,376,860	39.33
`ward_name`	496,716	14.19
`price`	218,494	6.24
`district_name`	101,933	2.91
`area`	4	<0.01

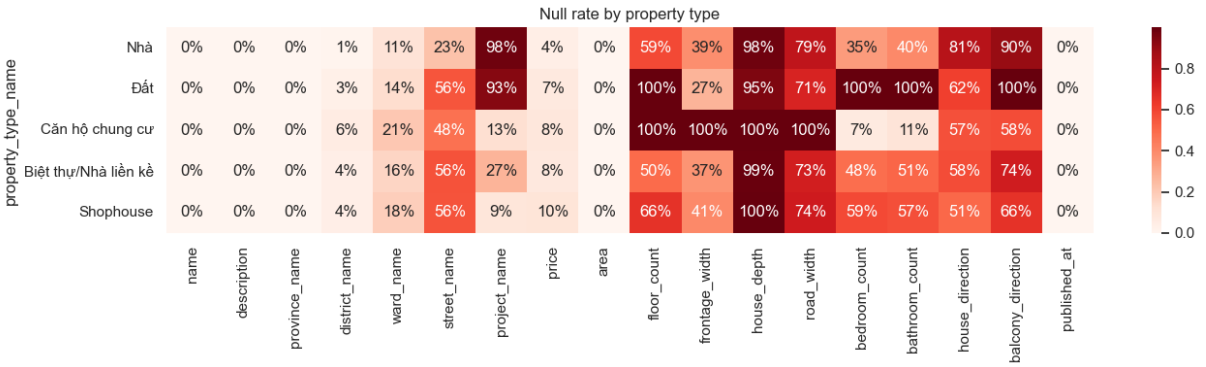


Figure 1: Null rate of each variable by property type.

Missingness differs by property type (Figure 1), indicating that absent attributes often reflect the nature of the property:

- Apartments (*Căn hộ chung cư*): `floor_count`, `frontage_width`, `house_depth`, and `road_width` are missing in 100% of the listings.
- Land (*Đất*): `floor_count`, `bedroom_count`, `bathroom_count`, and `balcony_direction` are missing in 100% of the listings.
- `project_name` is missing in 98% of houses (*Nhà*) and 93% of land listings, compared with 13% of apartments. A missing `project_name` often indicates that the listing does not belong to a named development project.
- `price` is missing in 4% to 10% of the listings of each type.

Rows with missing values were retained in the dataset, and listings without a valid price are excluded later from the price analysis.

4.1.3 Validity Checks

Basic numeric validity checks identified the following cases:

- `price`: 57,320 listings have a price equal to zero, and no listing has a negative price.
- `area`: no listing has an area less than or equal to zero.
- `floor_count`, `bedroom_count`, `bathroom_count`: no negative values.

Listings without a valid price or area represent 7.88% of the raw data. This share ranges from 5.2% (*Nhà*) to 13.6% (*Shophouse*) across the five property types, and from 4.1% (Khánh Hòa and Lâm Đồng) to 18.8% (Hòa Bình) among the 15 provinces with the most listings. The variation in exclusion rates means that filtering does not affect every property type and province equally.

4.1.4 Temporal, Geographic and Direction Checks

Time, location, and direction variables require the following treatment:

- `published_at`: no missing values in the raw data, but 22,024 values could not be converted to a timestamp (invalid values were coerced to missing). The parsed values span 1 June 2025 to 30 March 2026.
- Geography: the data contain 63 provinces and 627 districts. A total of 15 district names appear in more than one province, so `district_name` should be used together with `province_name`.
- Direction: `house_direction` and `balcony_direction` are free text, with 2,414,214 and 2,933,342 missing values, respectively. The same direction is sometimes written in two notations (for example, *Đông Nam* and *Đông - Nam*), and a few entries contain several directions, each occurring at most five times in `house_direction` and at most four times in `balcony_direction`. No standardisation was applied at this stage.

4.1.5 Duplicate Analysis

Listings were flagged as duplicates when they share the same normalised text of `name`, `province_name`, `district_name`, and `ward_name` (lower case, diacritics removed, *đ* mapped to *d*, whitespace collapsed), together with the same `area` and `price`, as an earlier listing. The first occurrence is kept.

Under this definition, 865,362 listings (24.72% of the raw data) are duplicates. The ten most repeated combinations occur between 60 and 121 times each, all in Hồ Chí Minh, and eight of the ten share the same title opening (*xuat canh ban gap*, in normalised form). The most repeated combination is a house on Lê Thánh Tôn street (Bến Nghé ward, District 1) of about 43 m^2, priced at 15.5 billion VND, which appears 121 times.

4.1.6 Extreme Value Analysis

Price spans several orders of magnitude even after invalid values and duplicate listings are removed (Figure 2).

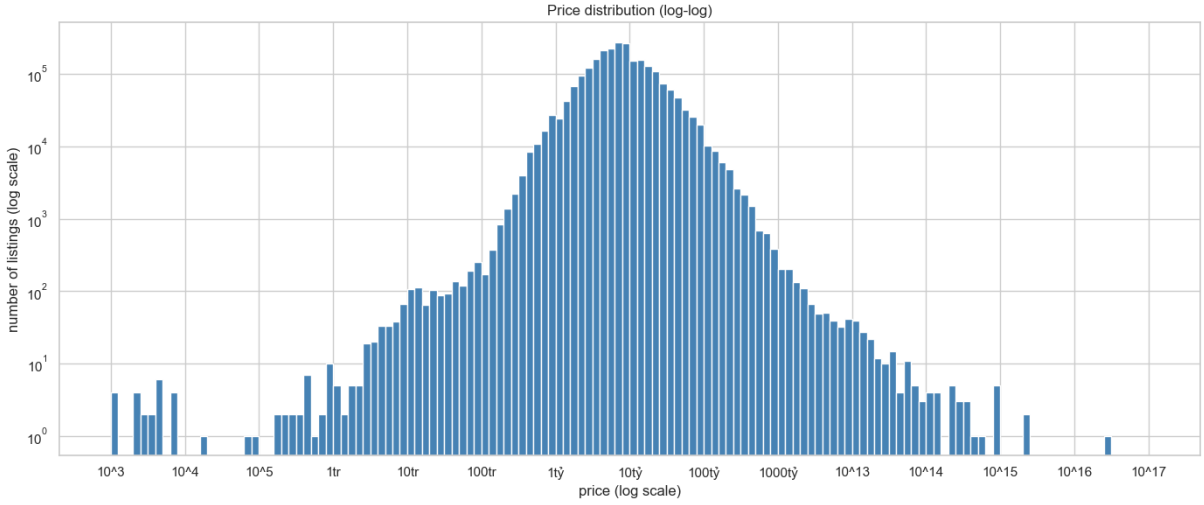


Figure 2: Distribution of asking price after removing invalid prices and duplicates (logarithmic scale on both axes).

Figure 2 shows that most listings are concentrated between about 10^9 and 10^{10} VND, with isolated listings below 10^4 VND and above 10^{15} VND. The ten lowest and ten highest prices were inspected:

- The ten lowest prices are between 1,000 and 3,000 VND. In several of these listings the title states a different price; for example, a listing recorded at 2,000 VND is titled as 1.6 billion VND.
- The ten highest prices range from about 4.49×10^{14} to 2.66×10^{16} VND, and some of these listings have very large areas (for example, 999,999 m^2).

These observations suggest data entry errors, for example in the unit of the price. A range of $[10^8, 10^{12}]$ VND (100 million to 1,000 billion VND) was therefore examined: it would remove 2,574 listings (0.107%), and it was applied as a cleaning step (Section 4.2).

Similar extremes occur in physical attributes and price per square metre (Figure 3).

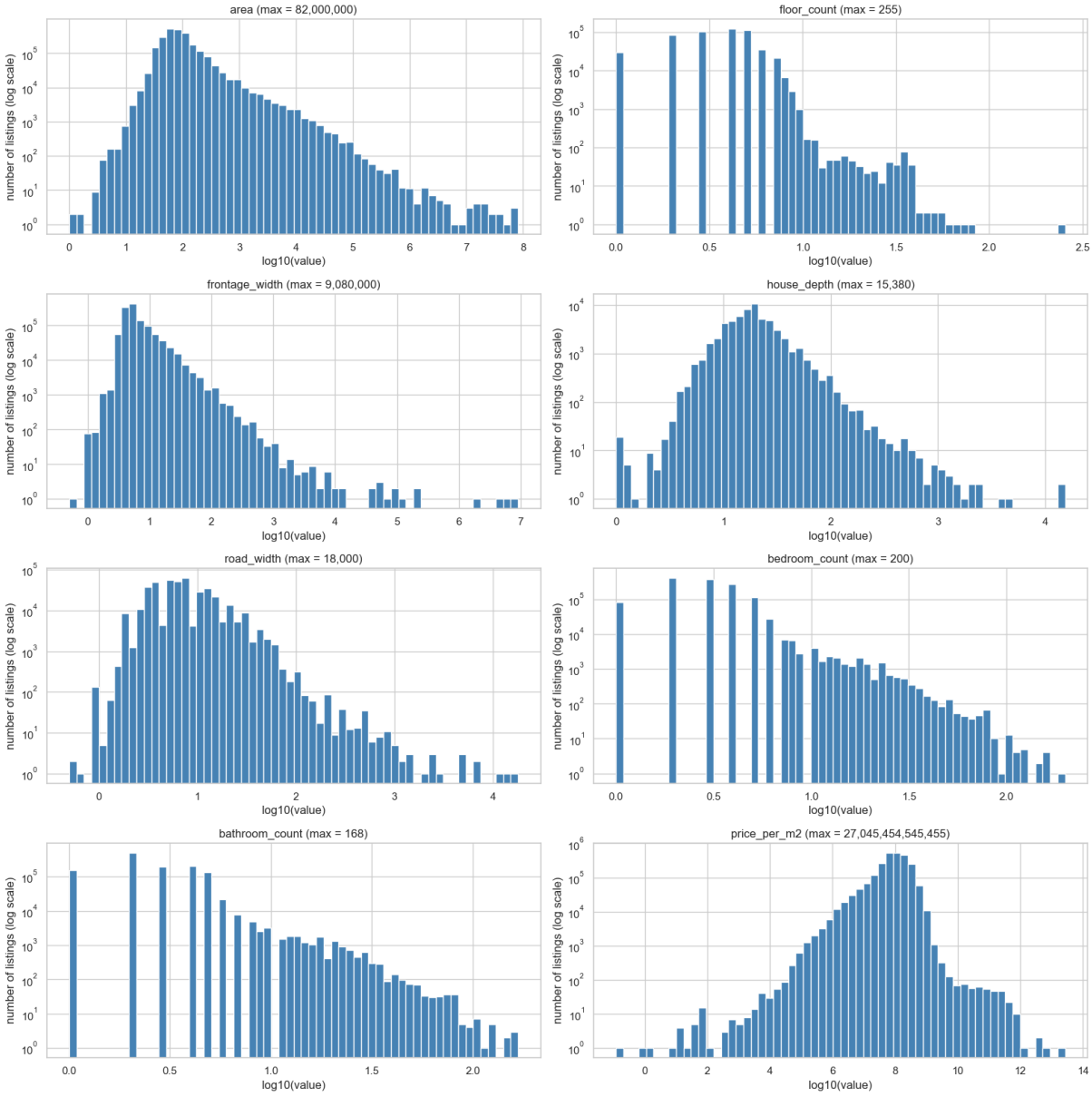


Figure 3: Distributions of the numeric attributes and of the price per square metre ($\log_{10}$ scale; vertical axis in logarithmic scale), before the plausibility limits are applied.

Figure 3 shows narrow central ranges and sparse tails extending over several orders of magnitude. Examples include areas up to 82 million m^2, frontage widths above 9 million m, and floor counts up to 255. These observations motivate the plausibility limits in Table 5. Invalid optional attributes are set to missing, allowing other information from the listing to remain available for modelling.

4.2 Analytical Dataset Construction

The analytical dataset was constructed using the six filtering steps in Table 6. Listings first require positive price and area; flagged duplicates and prices outside $[10^8, 10^{12}]$ VND are then removed.

The remaining steps remove listings without a valid publication date, apply plausibility limits to `area` and the other numeric attributes (Table 5), and remove listings with a price per square metre above 2 billion VND. For `area`, listings outside the limits were removed. For the other six attributes, the out-of-range values were set to missing and the listings were retained. The number of values outside the limits was counted before the area filter was applied. The final dataset contains 2,397,001 listings.

Table 5: Plausibility limits applied to the numeric attributes.

Variable	Valid range	Values outside (n)	Treatment
`area` (m^2)	$[10, 100{,}000]$	876	Row removed
`floor_count`	$[1, 50]$	8	Set to missing
`frontage_width` (m)	$[1, 200]$	761	Set to missing
`house_depth` (m)	$[2, 500]$	70	Set to missing
`road_width` (m)	$[1, 100]$	424	Set to missing
`bedroom_count`	$[1, 50]$	340	Set to missing
`bathroom_count`	$[1, 50]$	267	Set to missing

Table 6: Number of listings after each filtering step.

Step	Rule	Listings (n)	Removed (n)
0	Raw data	3,500,744	—
1	Price > 0 and area > 0	3,224,930	275,814
2	Duplicates removed	2,415,372	809,558
3	Price within $[10^8, 10^{12}]$ VND	2,412,798	2,574
4	Valid publication date	2,398,343	14,455
5	Area within $[10, 100{,}000]$ m^2	2,397,467	876
6	Price per $m^2 \leq 2 \times 10^9$ VND (final)	2,397,001	466

The number removed in step 2 is the difference between the counts before and after the step.

4.3 Price Analysis

The following results describe the 2,397,001 retained listings.

4.3.1 General Dataset Profile

Table 7 reports descriptive statistics of the numeric variables. The main observations are:

- `price` and `area` are available for all listings, whereas the other attributes are available for fewer listings (from 58,878 for `house_depth` to 1,304,397 for `bedroom_count`).
- The median price is 7.4 billion VND and the mean (1.49×10^{10} VND) is about twice as large, which indicates a right-skewed distribution.
- `area` behaves similarly: the median is 80 m^2, the mean is 217 m^2, and the standard deviation is 1,619 m^2.
- The maximum of each attribute equals the upper limit in Table 5 (for example, 100,000 m^2 for `area` and 200 m for `frontage_width`), so the tails are bounded.

Table 7: Numeric-variable summary for the analytical dataset (N = 2,397,001). Prices are in VND; dimensions use the units stated in Table 2.

Variable	Available (n)	Mean	Std. dev.	Median	Maximum
Price	2,397,001	1.49×10^{10}	3.20×10^{10}	7.40×10^9	10^{12}
Area	2,397,001	217.38	1,619.32	80.00	100,000
Floor count	516,628	3.90	1.76	4	50
Frontage width	1,208,300	7.54	8.97	5.00	200
House depth	58,878	22.57	20.84	18.75	500
Road width	414,645	9.08	8.62	6.00	100
Bedroom count	1,304,397	3.30	2.34	3	50
Bathroom count	1,220,972	3.05	2.34	2	50

4.3.2 Distribution of Price and Area

Both price and area remain right-skewed after cleaning; logarithmic axes make their central distributions easier to inspect.

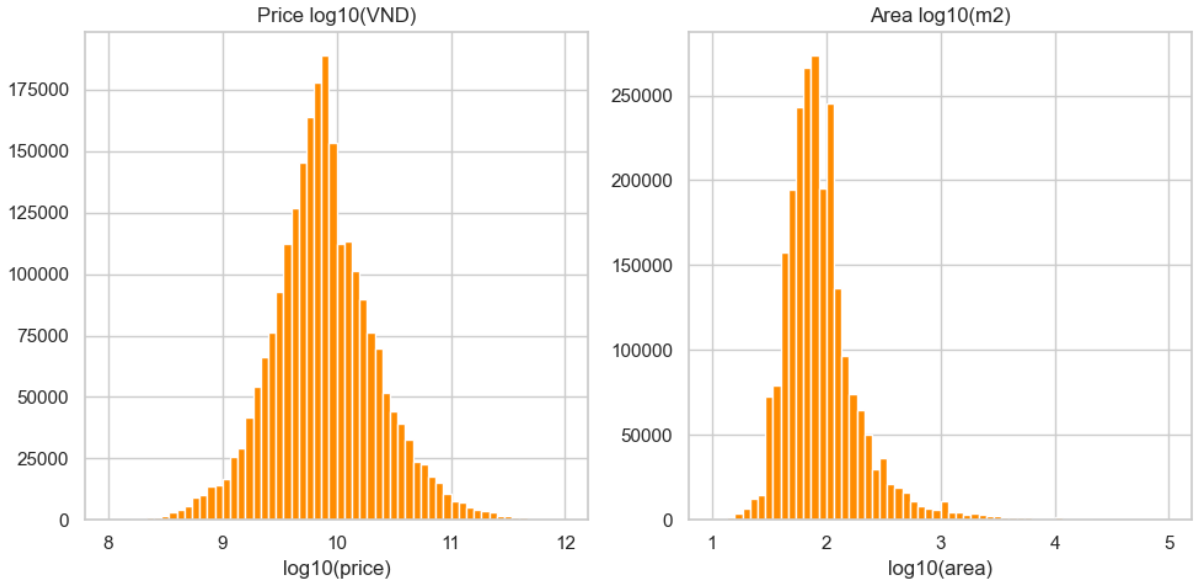


Figure 4: Histograms of $\log_{10}$(price) (left) and $\log_{10}$(area) (right).

Figure 4 shows that the distribution of price on the logarithmic scale is unimodal and approximately symmetric, with a mode at about 9.9 (roughly 8 billion VND) and values mainly between 8.5 and 11.5. The distribution of area is unimodal with a mode at about 1.9 to 2.0 (roughly 80 to 100 m^2) and a right tail that extends to about 5 (the upper limit of 100,000 m^2). Given the right skew shown in Table 7, this supports the use of a log transformation for price.

4.3.3 Group Coverage

A threshold of 100 listings is used to describe group coverage. Groups below this threshold have limited support for separate empirical comparisons.

- Property types: all 5 types have more than 100 listings: houses (1,074,788), land (560,294), apartments (544,268), villas/townhouses (178,654), and shophouses (38,997).
- Provinces: 60 of the 63 provinces have at least 100 listings.
- Listings are concentrated in Hồ Chí Minh (928,687) and Hà Nội (776,760), which together account for about 71% of the analytical dataset.

4.3.4 Area–Price Relationship

A random sample of 20,000 listings (seed 42) illustrates the area–price relationship without plotting the full dataset.

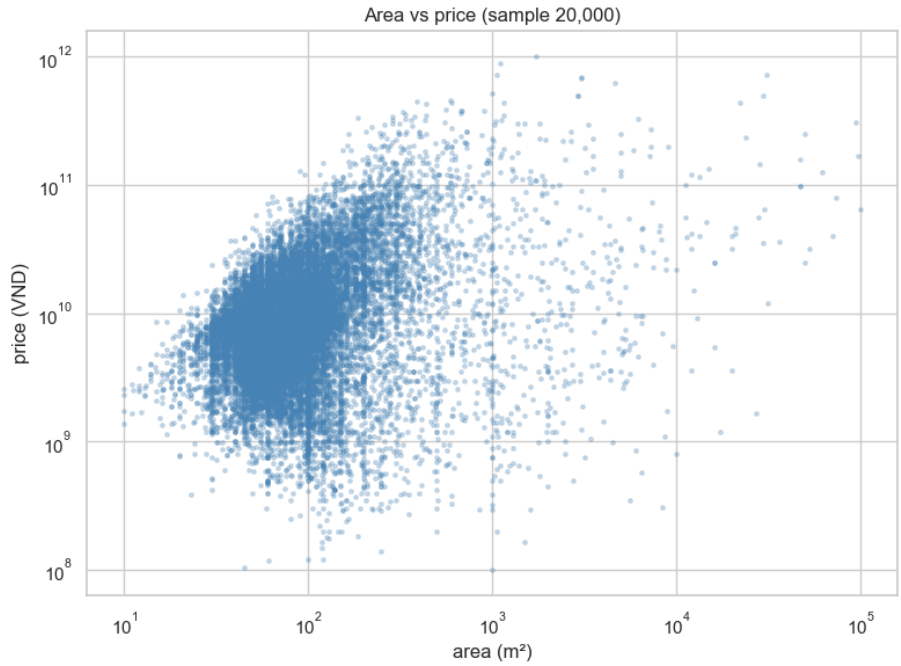


Figure 5: Area versus asking price on logarithmic axes (random sample of 20,000 listings).

Figure 5 shows that the area axis spans 10 to 10^5 m^2, the limits applied in the cleaning. Most points lie at areas of roughly 30 to 300 m^2 and prices of roughly 10^9 to 10^{10} VND. The cloud shows an upward tendency, so larger areas are generally associated with higher prices, but the dispersion at a given area is wide, and points become sparse above 10^3 m^2. The wide dispersion motivates including location and property characteristics alongside area in the prediction models.

4.3.5 Price per Square Metre by Property Type

The price per square metre (`price_per_m2`) is `price` divided by `area`. Its $\log_{10}$ distribution was plotted for each of the five property types, with the horizontal axis restricted to 6–10.

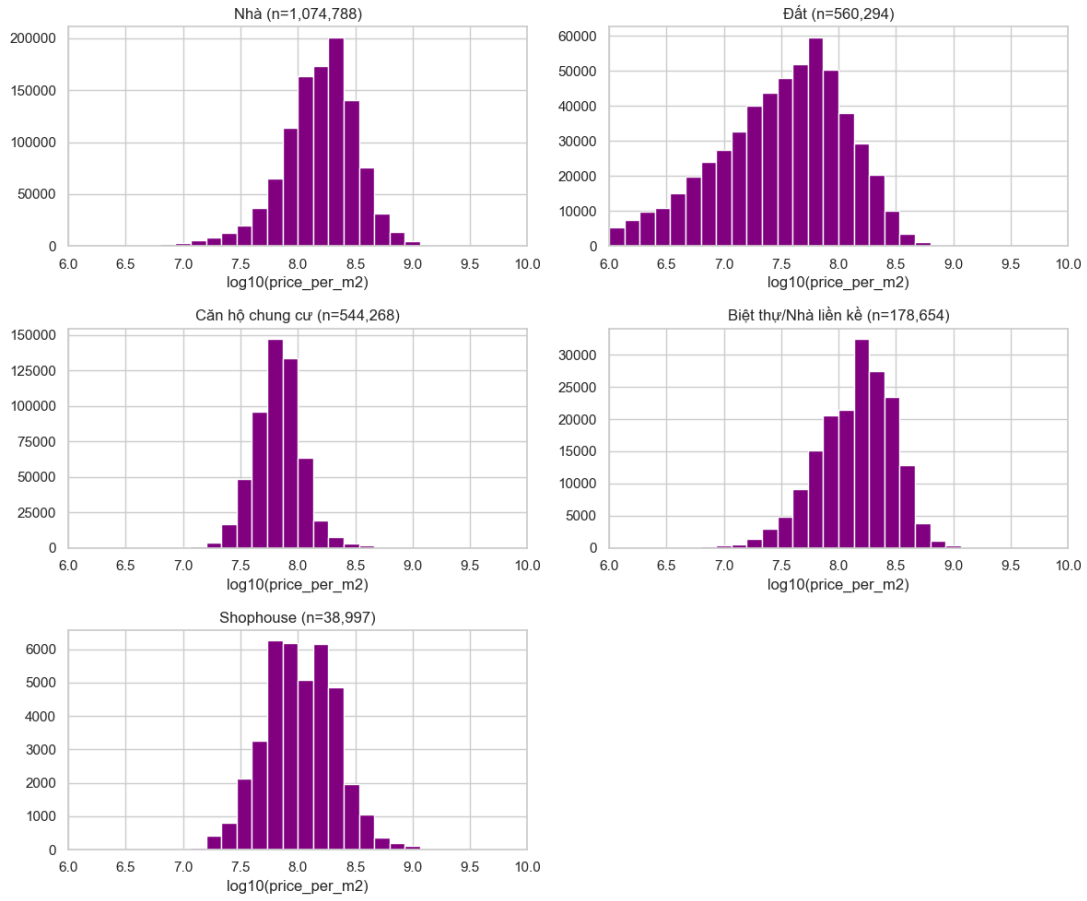


Figure 6: Distribution of log price per square metre across the five property types in the analytical dataset.

Figure 6 shows that the distributions differ in location and spread across property types (the data are capped at 2 billion VND per m^2, i.e. $\log_{10} \approx 9.3$):

- Houses and villas/townhouses are centred at higher values (modes at about 8.3 and 8.2, roughly 200 and 160 million VND per m^2) than apartments and land (modes at about 7.8).
- Land has the widest distribution, skewed to the left, and apartments have the narrowest.
- Shophouses show two local maxima, at about 7.8–7.9 and 8.2. This may indicate a mix of different kinds of listing (for example, sale and rental), but it was not tested and remains a hypothesis.

4.4 Summary of Findings

- The raw data contain 3,500,744 listings. Missing values are frequent for structural attributes and depend on the property type.
- 24.72% of the raw listings were flagged as duplicates and 7.88% lack a valid price or area. After all filtering steps, 2,397,001 listings remain.
- Price and area are right-skewed, and the log transformation of price gives an approximately symmetric distribution. Extreme values were handled with fixed plausibility limits.
- All five property types and 60 of 63 provinces have sufficient listings, although listings are concentrated in two provinces. Area and price show a positive but widely dispersed relationship, and the price per square metre differs by property type.

5 Model Selection and Training

5.1 Machine Learning

The machine learning experiments predict property asking prices from physical attributes, location, property category, and advertising text. The experimental design progresses from constant and local-median baselines to regularised linear regression, tree ensembles, and models combining textual and structured information. This progression assesses whether nonlinear interactions and listing descriptions contribute predictive information beyond a linear representation of property attributes.

Figure 7 summarises the modelling workflow. All models share the same chronological training, validation, and test partitions. Learned preprocessing is fitted on training observations during model selection; subsequent refitting procedures are specified separately for each model family.

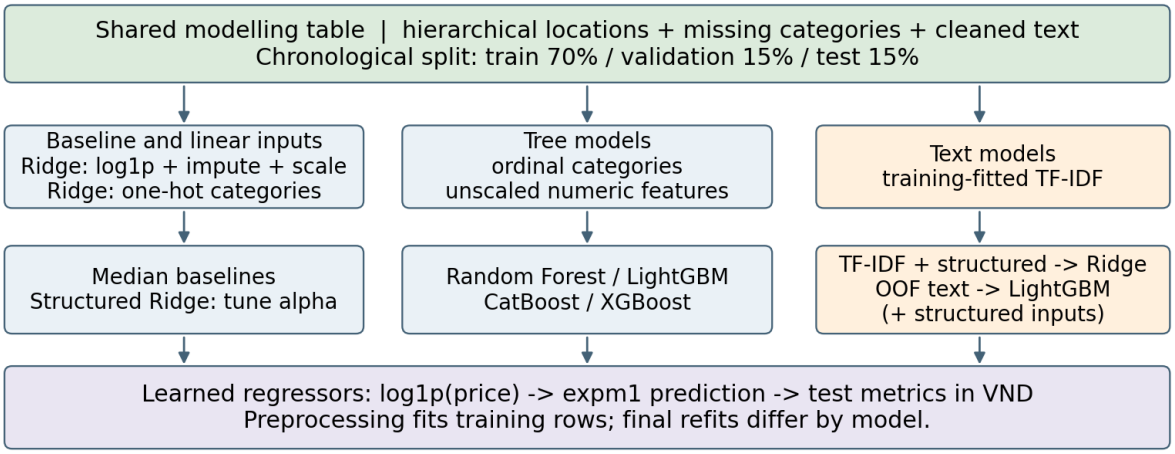


Figure 7: Machine learning workflow. Median baselines predict prices directly; learned regressors predict log price. OOF denotes the out-of-fold text predictions used to train the text-enhanced LightGBM.

5.1.1 Feature Representation and Training Protocol

Structured predictors. The structured representation contains seven numeric and eight categorical variables (Table 8). The target is the asking price in Vietnamese dong (VND). Publication time determines the chronological partitions and is excluded from the predictor matrix. A listing's own price per square metre and other target-derived quantities are also excluded from ordinary predictors.

Table 8: Structured features used by the machine learning models.

Feature group	Variables
Numeric	Area, floor count, frontage width, house depth, road width, bedroom count, bathroom count
Categorical	Property type, province, district, ward, street, development project, house direction, balcony direction

Location categories are represented hierarchically: district names are prefixed with their province, while ward and street names are prefixed with the resulting district key. This distinguishes identical local names in different administrative areas. Missing categorical values receive a dedicated category. Numeric missing values are retained for model-specific treatment.

Text representation. The title and description are concatenated, lowercased, and stripped of the phone-number placeholder. To reduce direct target disclosure, punctuation-delimited clauses containing monetary terms, including *giá*, *tỷ*, *triệu*, VND, and USD, are removed. Tokens containing digits are subsequently removed, and whitespace is normalised. This conservative processing also removes useful

numeric descriptions, such as area and room counts; these remain available through the structured predictors.

Chronological evaluation. The 2,397,001 prepared listings are ordered by publication time. The oldest 70% form the training partition (1,677,900 observations), the next 15% form validation (359,550), and the most recent 15% form testing (359,551). This evaluates prediction on later listings using earlier observations. The test partition is excluded from fitting preprocessing transformations, selecting hyperparameters, and determining stopping points.

Target transformation. The median baselines predict original-scale prices. All learned regressors use the natural-log target

$$z_i = \log(1 + y_i), \qquad \widehat{y}_i = \exp(\widehat{z}_i) - 1, \tag{1}$$

where y_i denotes asking price in VND. The transformation compresses the upper price tail during fitting. Predictions are transformed back before calculating original-scale evaluation metrics. Because log-price loss and price loss weight errors differently, validation criteria are identified explicitly below.

5.1.2 Baseline Models and Ridge Regression

Median-price baseline. The first baseline assigns every property the median training price:

$$\widehat{y}_i = \text{median}_{j \in \mathcal{T}}(y_j), \tag{2}$$

where $\mathcal{T}$ denotes the training partition. This constant predictor provides an absolute-error benchmark without using property attributes.

Local median price per square metre. The second baseline uses area, district, and property type. For each district–property-type group g, the median price per square metre is estimated from training observations:

$$m_g = \text{median}_{j \in \mathcal{T}: g_j = g} \left(\frac{y_j}{a_j} \right), \qquad \widehat{y}_i = a_i m_{g_i}, \tag{3}$$

where a_i is area. A group absent from training uses the overall training median price per square metre. Only training targets contribute to these local pricing estimates.

Ridge regression. Ridge regression introduces an L_2 penalty to stabilise linear coefficient estimation [HK70]. It serves as the regularised linear benchmark for the structured predictors. Numeric variables undergo a logarithmic transformation, median imputation, addition of missingness indicators, and standardisation. Categorical variables are one-hot encoded, pooling levels observed fewer than 50 times. Unseen categories use the pooled category when available. Candidate preprocessing statistics are estimated exclusively from training rows.

The regression objective is

$$\min_{b, \boldsymbol{\beta}} \sum_{i \in \mathcal{T}} \left(z_i - b - \mathbf{x}_i^\top \boldsymbol{\beta} \right)^2 + \alpha \|\boldsymbol{\beta}\|_2^2, \tag{4}$$

where α controls coefficient shrinkage [HK70]. Three values, $\alpha \in \{1, 10, 100\}$, are compared using validation MAE after inversion to VND. The selected value is $\alpha = 1$. The entire preprocessing and regression pipeline is subsequently refitted on training plus validation observations before test prediction.

Figure 8 illustrates the fitted model's test predictions. Points close to the diagonal indicate agreement between actual and predicted prices. Several expensive properties fall below the diagonal, indicating underprediction in the upper price range.

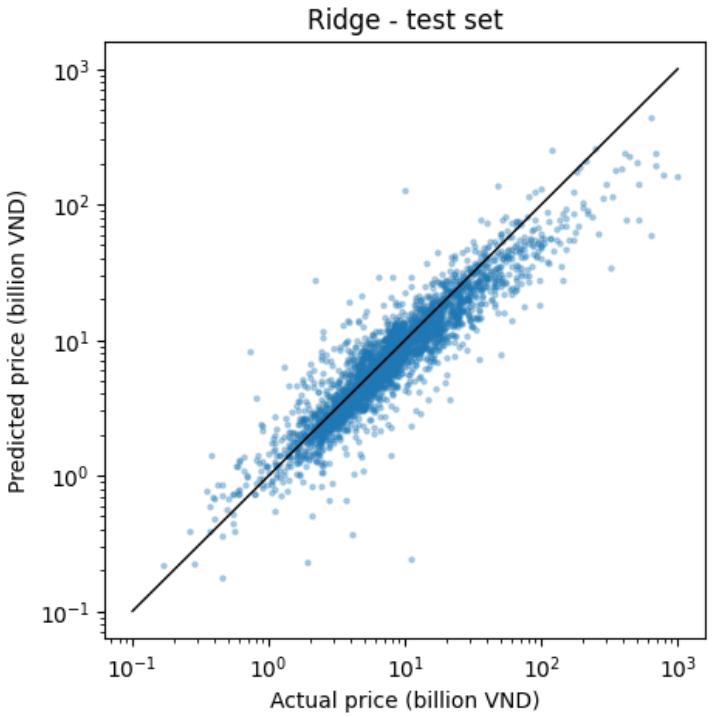


Figure 8: Actual and predicted asking prices for 3,000 randomly selected test listings (seed 42) using structured Ridge. Both axes are logarithmic and measured in billion VND; the diagonal denotes exact prediction.

5.1.3 Tree Ensemble Models

Random Forest aggregates randomised regression trees [Bre01], while gradient boosting constructs an additive predictor through successive loss-reducing updates [Fri01]. These complementary ensemble approaches allow nonlinear relationships and interactions between physical attributes, property type, and location.

Input encoding. Categorical levels are mapped to integer codes using a training-fitted encoder; unseen levels receive code -1. Numeric features retain their original scale and missing values. Random Forest treats category codes as numeric inputs. LightGBM and CatBoost receive explicit categorical-column declarations, while XGBoost receives categorical-typed columns with categorical support enabled. Consequently, the integer encoding does not imply the same treatment of categories across all four models.

Random Forest. Following the randomised-tree aggregation framework of Breiman [Bre01], the forest contains 100 regression trees, with at least five observations per leaf and random seed 42. Each tree draws a bootstrap sample containing 10% of the fitting population to reduce the training cost per tree.

The search crosses maximum depth $\{12, 20\}$ with feature fraction $\{0.5, 1.0\}$, producing four candidate configurations. Candidates fit training log prices and are ranked by validation MAE in VND. Maximum depth 20 and feature fraction 1.0 are selected. The final forest is refitted on training plus validation observations, while the categorical mapping remains fitted on training data.

LightGBM. LightGBM is a scalable gradient-boosted decision-tree framework [Ke+17]. The model uses up to 2,000 trees, learning rate 0.05, 63 leaves, and seed 42. Validation log-MAE and the default log-MSE are monitored, with early-stopping patience of 50 rounds. Categorical columns are declared explicitly, and numerical features are not standardised.

Figure 9 ranks district and area highest by accumulated split gain [Fri01]. The model therefore relies strongly on local geography and property size; gain measures predictive use rather than a causal price effect.

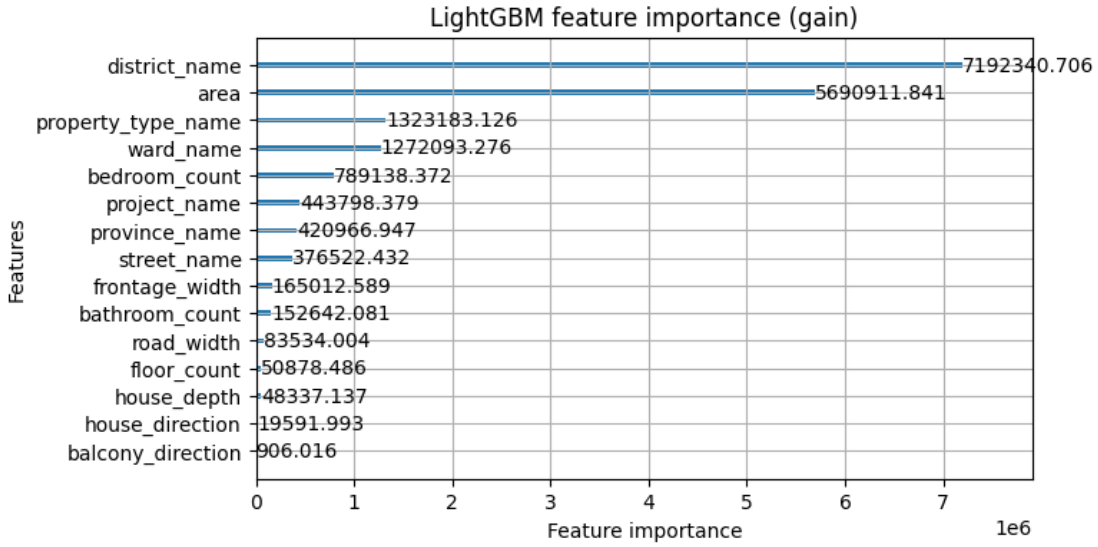


Figure 9: LightGBM feature importance by accumulated split gain. District categories include the province prefix.

CatBoost. CatBoost provides gradient boosting with specialised treatment of categorical features [Pro+18]. The fitted configuration allows 2,000 iterations with learning rate 0.1, maximum categorical-combination complexity one, 64 numerical quantisation borders, and seed 42. Categorical columns are explicitly identified in the integer-encoded feature table. The training loss is RMSE on log price, while validation log-MAE determines checkpoint selection with early-stopping patience of 30 rounds.

XGBoost. XGBoost is a scalable tree-boosting system developed by Chen and Guestrin [CG16]. The configuration uses up to 2,000 trees, learning rate 0.05, maximum depth eight, histogram-based tree construction, categorical support, parallel CPU execution, and seed 42. Training operates on log price, monitors validation log-MAE, and uses early-stopping patience of 50 rounds.

The boosting configurations are fixed experimental settings rather than the outcome of a parameter-grid search. All three reach the 2,000-round limit in the recorded training runs. Early stopping is therefore configured but does not shorten those runs. These models retain their training-fitted, validation-monitored predictors rather than undergoing a subsequent training-plus-validation refit.

Table 9: Training configurations and selection procedures for structured regressors.

Model	Main settings	Selection and final fitting
Ridge	$\alpha \in \{1, 10, 100\}$	Validation MAE in VND; $\alpha = 1$; refit on train + validation
Random Forest	100 trees; depth $\{12, 20\}$; feature fraction $\{0.5, 1.0\}$; leaf size 5	Validation MAE in VND; depth 20 and fraction 1.0; refit on train + validation
LightGBM	2,000 trees; learning rate 0.05; 63 leaves	Validation log-price metrics; patience 50; training-fitted model
CatBoost	2,000 iterations; learning rate 0.1; categorical-combination complexity 1; 64 borders	Validation log-MAE; patience 30; training-fitted model
XGBoost	2,000 trees; learning rate 0.05; depth 8; histogram method	Validation log-MAE; patience 50; training-fitted model

5.1.4 Combining Text and Structured Features

Textual information is incorporated using two complementary designs: a joint sparse representation fitted by Ridge, and a text-derived prediction used as an additional LightGBM input. Both preserve the

structured attributes while introducing information expressed in the listing description.

TF-IDF representation. Term frequency–inverse document frequency (TF-IDF) assigns weights according to within-document term occurrence and corpus-level document frequency, following established term-weighting methods [SB88]. The representation uses word unigrams, minimum document frequency five, a maximum vocabulary of 50,000 terms, sublinear term frequency, and 32-bit floating-point storage. The resulting vocabulary contains 23,233 terms. Vocabulary selection and IDF estimation use training documents only; validation and test documents reuse the fitted representation.

Ridge on combined features. The TF-IDF matrix is horizontally concatenated with the preprocessed structured matrix, yielding 29,592 columns. Sparse storage avoids materialising a large dense design matrix. Ridge regularisation [HK70] is selected from $\{1, 10, 100\}$ using validation MAE in VND, with $\alpha = 1$ selected. The final regression coefficients are fitted on the combined training and validation matrices. The vocabulary and structured preprocessing remain fitted on training observations, in contrast to the full-pipeline refit used for structured-only Ridge.

Figure 10 visualises the validation searches. Both Ridge variants select $\alpha = 1$, and the joint representation has lower validation MAE at all three tested values. The Random Forest comparison supports maximum depth 20 and feature fraction 1.0. These are model-selection results, rather than test-set performance estimates.

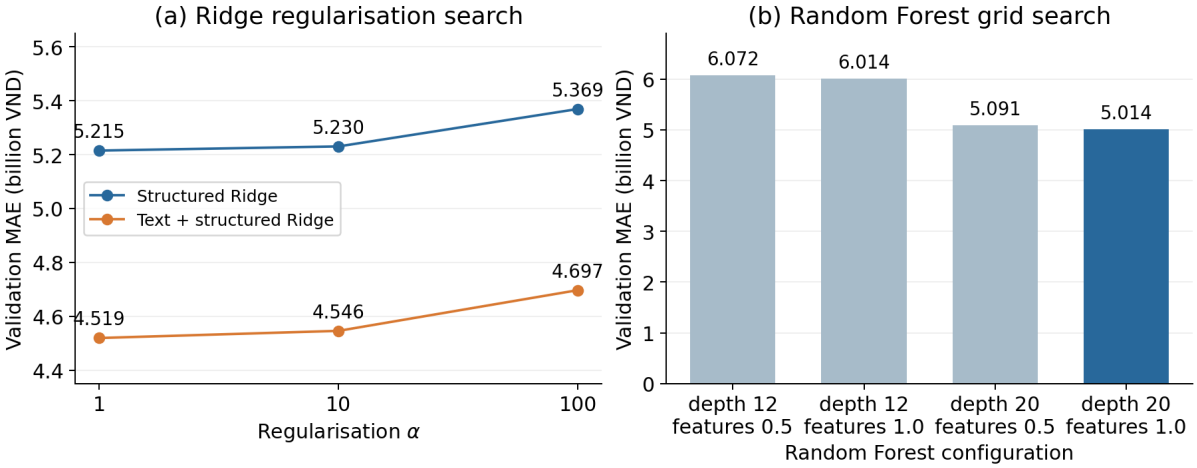


Figure 10: Validation MAE for (a) the two Ridge representations and (b) four Random Forest configurations. Lower values are better; the darker bar identifies the selected forest.

LightGBM with a text-derived prediction. The second design follows the prediction-as-feature principle of stacked generalisation [Wol92]. A text-only Ridge model predicts log price, and that prediction is appended to the 15 structured attributes as a sixteenth input to LightGBM. To avoid using in-sample fitted target estimates as training features, five-fold cross-validation produces out-of-fold text predictions: each fold is predicted by a Ridge model fitted on the other four folds.

A text Ridge model is then fitted on all training documents to produce validation and test features. Its regularisation parameter reuses the value selected for the joint Ridge model. LightGBM [Ke+17] is trained on the augmented feature matrix with the same 2,000-tree limit, learning rate 0.05, 63 leaves, seed 42, and early-stopping patience of 50 rounds as the structured model.

Five ordinary folds are formed within the chronological training partition. Ridge coefficients are estimated separately for each fold, while the vocabulary remains fitted on the full training partition.

5.2 Deep Learning

The neural models comprise a tabular MLP with categorical embeddings and a pretrained Vietnamese text encoder. Boosted trees provide the tabular comparison, following the emphasis on strong baselines in Gorishniy et al. [Gor+21].

Both use the chronological evaluation protocol in Section 5.1.1. Figure 11 contrasts their input representations and regression heads.

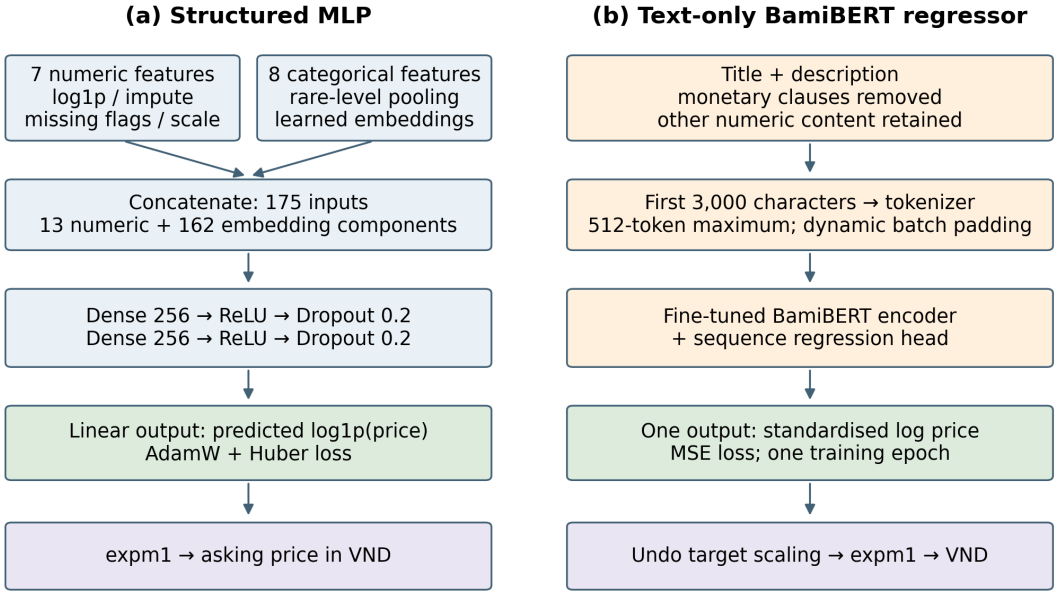


Figure 11: Neural regression architectures: (a) the structured MLP with categorical embeddings and (b) the text-only BamiBERT regressor. Both invert their target transformations to predict VND prices.

5.2.1 Tabular MLP with Categorical Embeddings

Inputs and embeddings. The numeric branch applies log transformation, median imputation, missingness indicators, and standardisation using training-fitted statistics. Categorical levels observed fewer than five times are pooled, and unseen levels receive a dedicated code. Each categorical variable has a jointly learned embedding, as in neural tabular prediction [Gor+21].

For a categorical variable with encoded cardinality c, the embedding dimension is set by the experimental heuristic

$$d(c) = \min\left(32, \operatorname{round}\left(1.6c^{0.56}\right)\right). \tag{5}$$

The eight embeddings contribute 162 components. Seven numeric attributes and six missingness indicators contribute another 13, giving 175 inputs. Area has no missing training values and therefore receives no missingness indicator.

Network and objective. Two 256-unit hidden layers use rectified linear activations [GBB11] and dropout probability 0.2 [Sri+14]. A linear output predicts log price, with its bias initialised to the mean training log price.

Training minimises Huber loss [Hub64] on the residual $r_i = \widehat{z}_i - z_i$:

$$\ell_\delta(r_i) = \begin{cases} \frac{1}{2}r_i^2, & |r_i| \le \delta, \\ \delta\left(|r_i| - \frac{1}{2}\delta\right), & |r_i| > \delta, \end{cases} \qquad \delta = 1. \tag{6}$$

Huber loss combines quadratic sensitivity near zero with linear tails for large errors. It is applied to log-price residuals.

Optimisation and stopping. The optimiser is AdamW, which decouples weight decay from adaptive gradient updates [LH19]. The learning rate is 10^{-3}, weight decay is 10^{-4}, batch size is 1,024, and the random seed is 42. Training is limited to 30 epochs, with validation log-MAE evaluated after every epoch. Early stopping triggers after three consecutive epochs without improvement, and the best weights are restored.

Validation log-MAE reaches its minimum of approximately 0.2246 at epoch 10 (Figure 12). Training stops after epoch 13 and restores epoch 10, without refitting on validation targets.

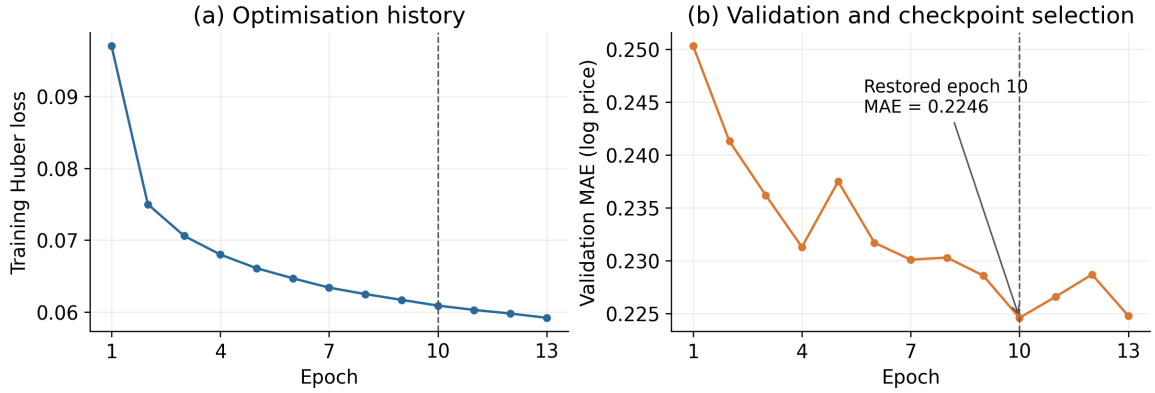


Figure 12: MLP training Huber loss and validation log-MAE. The dashed line marks the restored epoch-10 checkpoint. The panels show different objectives; values are rounded to four decimal places.

5.2.2 Fine-Tuning a Vietnamese Text Encoder

Representation and regression head. The text model fine-tunes the `Qualcomm-AI-Research/BamiBERT` Vietnamese checkpoint with a newly initialised scalar regression head. Its contextual representations use transformer self-attention [Vas+17], with task-specific fine-tuning following the bidirectional encoder paradigm [Dev+19].

Title and description text are lowercased and stripped of monetary clauses and phone placeholders. Unlike TF-IDF, other numbers are retained because this model has no separate numeric inputs. Documents are limited to 3,000 characters and 512 tokens, with dynamic padding within each batch.

Target scaling. Training log prices are standardised using their mean $\mu_{\mathcal{T}}$ and sample standard deviation $s_{\mathcal{T}}$:

Fine-tuning configuration. The experiment uses one epoch, learning rate 3×10^{-5}, warm-up over the first 6% of training steps, weight decay 0.01, training batch size 64, evaluation batch size 256, and seed 42. Mixed precision uses BF16 when supported and otherwise FP16 on CUDA. Evaluation and checkpoint saving occur at the end of each epoch, with checkpoint selection based on validation MSE.

End-of-epoch validation MSE is 0.10254. Figure 13 shows the training trajectory and this single validation measurement. The one-epoch limit determines training duration; an early-stopping callback with patience one is also configured.

5.2.3 Combining Structured and Textual Predictors

Stacked generalisation [Wol92] combines seven log-price predictors: LightGBM, CatBoost, XGBoost, MLP, joint text–structured Ridge, text-enhanced LightGBM, and BamiBERT. Predictions share the same observation indices and validation/test labels.

Four combinations distinguish the contribution of the text-based models:

1. **Structured:** LightGBM, CatBoost, XGBoost, and MLP.
2. **Structured + TF-IDF:** the structured combination plus joint Ridge and text-enhanced LightGBM.
3. **Structured + BamiBERT:** the structured combination plus the text encoder.
4. **All models:** all seven base predictors.

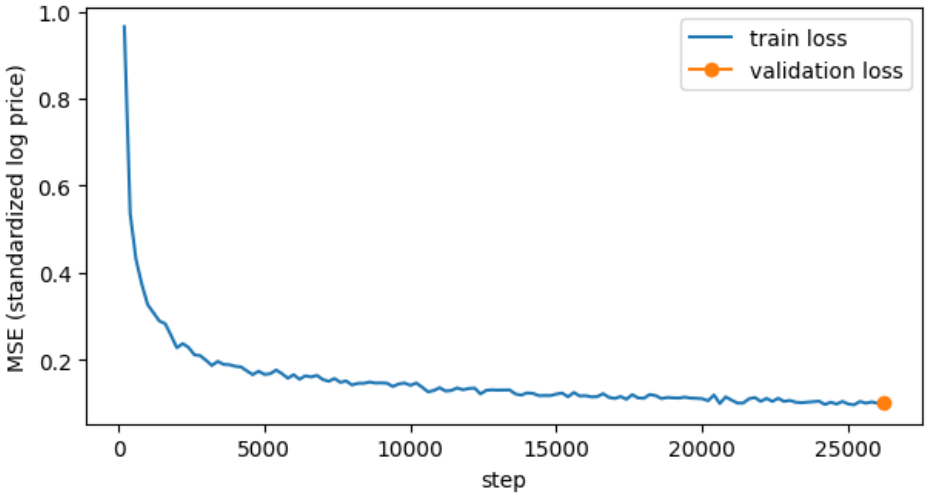


Figure 13: BamiBERT fine-tuning over one epoch. Training MSE is measured on standardised log price; the orange point denotes the single end-of-epoch validation measurement.

For each combination, coefficients are fitted on validation observations by nonnegative least squares with an intercept:

$$\min_{b,\,\mathbf{w}\geq 0} \sum_{i\in\mathcal{V}} \left(z_i - b - \sum_{m=1}^{M} w_m \widehat{z}_{im} \right)^2, \qquad \widehat{y}_i = \exp\left(b + \sum_{m=1}^{M} w_m \widehat{z}_{im} \right) - 1. \tag{7}$$

Here $\mathcal{V}$ is the validation partition. The coefficients are not constrained to sum to one, so the fitted predictor is a calibrated linear combination rather than a normalised weighted average. No test targets are used to estimate the coefficients.

The all-model blend assigns coefficients of approximately 0.41 to BamiBERT, 0.23 to text-enhanced LightGBM, and 0.19 to XGBoost (Figure 14). Smaller coefficients need not imply an uninformative predictor, because base predictions are correlated and displayed values are rounded.

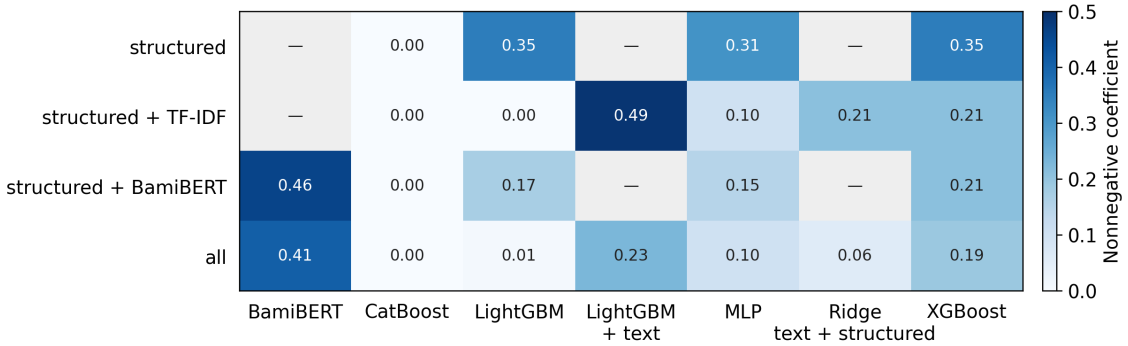


Figure 14: Nonnegative blend coefficients, rounded to two decimals. Dashes denote excluded predictors. A separate intercept is fitted, and coefficients are not constrained to sum to one.

The fitted coefficients are applied to test predictions without further adjustment. Evaluation results and the limits of the comparison are presented in Section 6.

6 Model Evaluation and Comparison

6.1 Evaluation Protocol

The 359,551 test listings follow training observations from June–December 2025 and validation observations from December 2025–January 2026. Testing spans January–March 2026. Test targets are excluded from hyperparameter fitting, stopping decisions, and blend estimation.

For n test observations, the four reported metrics are

$$\mathrm{MAE} = \frac{1}{n}\sum_{i=1}^{n}|y_i - \widehat{y}_i|, \qquad \mathrm{RMSE} = \sqrt{\frac{1}{n}\sum_{i=1}^{n}(y_i - \widehat{y}_i)^2}, \tag{8}$$

$$R^2 = 1 - \frac{\sum_i (y_i - \widehat{y}_i)^2}{\sum_i (y_i - \bar{y})^2}, \qquad \mathrm{MAPE} = \frac{100}{n}\sum_{i=1}^{n}\left|\frac{y_i - \widehat{y}_i}{y_i}\right|. \tag{9}$$

Here $\bar{y}$ is mean test price. MAE describes absolute error, RMSE emphasises large errors, R^2 compares squared error with a mean-price reference, and MAPE expresses relative error. MAE and RMSE are in billion VND; MAPE is a percentage. All targets are positive.

Structured Ridge and Random Forest are refitted on training plus validation. Joint Ridge refits its coefficients but retains training-fitted preprocessing; boosting and neural models retain their validation-monitored fits. For blending, joint Ridge supplies training-only validation predictions and refitted test predictions. Final fitting populations therefore differ.

6.2 Overall Predictive Performance

Table 10 compares 11 individual predictors and four blends. Local median pricing reduces MAE from the constant baseline's 11.700 to 7.225 billion VND, but increases RMSE to 42.901 billion VND. Its negative R^2 indicates worse squared error than the mean-price reference, despite the improvement in MAE.

Table 10: Held-out test performance for the evaluated models and blends. MAE and RMSE are in billion VND; bold marks the best observed values.

Predictor	MAE	RMSE	R^2	MAPE (%)
Median-price baseline	11.700	36.650	-0.059	98.29
Local median price per m^2	7.225	42.901	-0.451	47.35
Structured Ridge	5.730	23.865	0.551	31.97
Random Forest	5.406	22.063	0.616	31.22
LightGBM	4.464	18.189	0.739	26.24
CatBoost	5.242	21.065	0.650	31.66
XGBoost	4.485	18.401	0.733	26.42
Joint text–structured Ridge	4.935	21.606	0.632	28.09
LightGBM + text prediction	4.223	17.317	0.764	25.05
MLP + embeddings	4.620	19.264	0.707	26.83
BamiBERT (text only)†	4.355	17.353	0.763	26.86
Structured blend	4.241	17.357	0.762	25.90
Structured + TF-IDF blend	4.025	17.020	0.772	24.31
Structured + BamiBERT blend†	3.699	16.283	0.791	21.94
All-model blend†	**3.659**	**16.171**	**0.794**	**21.70**

† Results involving BamiBERT await checkpoint-reload verification and do not establish that all trained weights were restored correctly.

On this held-out test set, LightGBM has the lowest MAE among the evaluated purely structured individual predictors (4.464 billion VND), closely followed by XGBoost (4.485). Both have lower test MAE than the MLP.

6.3 Contribution of Text and Prediction Blending

On the held-out test set, adding TF-IDF to structured Ridge reduces MAE by 13.9%, from 5.730 to 4.935 billion VND. Adding a text-derived prediction to LightGBM gives a 5.4% reduction, from 4.464 to 4.223 billion VND, the lowest observed test MAE among the evaluated individual predictors.

BamiBERT achieves MAE of 4.355 billion VND from text alone. However, it retains non-price numbers while TF-IDF removes digit-bearing tokens. This comparison therefore combines a change in encoder with a change in available input information.

On this test partition, the structured blend achieves MAE of 4.241 billion VND; adding TF-IDF yields 4.025 and adding BamiBERT yields 3.699. The all-model blend has the best observed values across the four metrics among the evaluated configurations, with MAE of 3.659 billion VND and $R^2 = 0.794$ (Figure 15). BamiBERT-dependent results await checkpoint-reload verification.

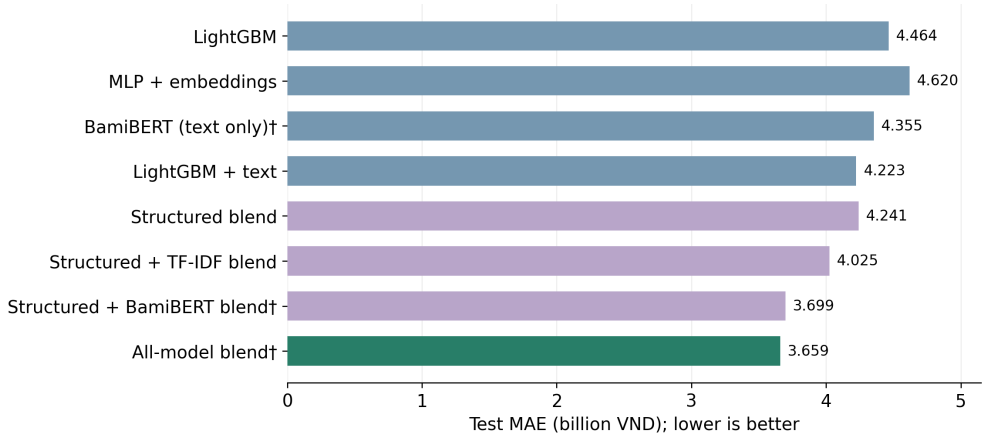


Figure 15: Held-out test MAE for selected predictors and blends; lower is better. † Results involving BamiBERT await checkpoint-reload verification.

The all-model blend reduces MAE by 13.4% relative to text-enhanced LightGBM and by 18.0% relative to structured LightGBM. Its gain over the smaller structured–BamiBERT blend is only 1.1%. Stacking combines complementary predictions [Wol92], but inference-time measurements are needed to assess the cost of this final increment in accuracy.

6.4 Limitations

Data quality and population coverage. The target is an advertised asking price, without verified transactions or reliable sale/rental labels. Ho Chi Minh City and Hanoi account for approximately 71% of the full analytical dataset. Inaccurate values may survive filtering, and revised titles or prices can leave repeated properties across partitions. Accuracy is not established separately for each province, property type, price range, or future period.

Model-selection and stacking protocol. Only one temporal holdout is evaluated, and final fitting populations differ. Validation is reused for tuning, stopping, and blend fitting. The text-derived feature uses ordinary five-fold prediction with a vocabulary fitted before the folds, rather than forward-chaining preprocessing. Repeated seeds, additional time windows, and uncertainty intervals are needed to assess stability and statistical significance.

Text leakage and checkpoint verification. Monetary-clause filtering can leave misspelled price expressions and numeric fragments, especially when other numbers are retained. BamiBERT checkpoint reload also reports missing and unexpected LayerNorm parameter names, distinct from a new regression head. The parameter mapping and residual price leakage require verification before stronger claims about BamiBERT or its blends. One training epoch does not establish the best performance attainable with longer fine-tuning.

7 Conclusion

This study examined machine learning and deep learning approaches to predicting real estate asking prices in Vietnam. Data-quality checks reduced 3,500,744 raw listings to 2,397,001 modelling observations. Hierarchical location features, missing-value treatment, text processing, and a chronological holdout provided a common basis for comparing the predictors.

On the held-out test set, LightGBM achieved lower MAE than the embedding-based MLP, consistent with the importance of competitive tabular baselines discussed by Gorishniy et al. [Gor+21]. Text-enhanced LightGBM reduced test MAE by approximately 5.4% relative to its structured counterpart. These comparisons describe the evaluated configurations under one chronological split, without isolating an intrinsic advantage of any model architecture.

Among the evaluated configurations, the all-model blend had the lowest observed test MAE (3.659 billion VND), with $R^2 = 0.794$. Because the blend includes BamiBERT, this result remains provisional pending checkpoint-reload verification. Its observed MAE was approximately 13.4% below the best evaluated individual predictor and 1.1% below the smaller structured–BamiBERT blend. The incremental gain should be assessed alongside inference cost once checkpoint correctness has been verified.

These findings apply to advertised asking prices under the study's evaluation protocol, subject to the data-quality and model-selection limitations in Section 6.4.

Further work should prioritise checkpoint and text audits, property-level identifiers, and repeated temporal evaluations. Province-, property-type-, and price-range analyses would reveal where errors are concentrated, while transaction-verified targets and uncertainty estimates would support a more realistic assessment of practical valuation performance.

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